CyberBrick RC Setup Guide

M1A2 Abrams SEP v3 — Illustrated desktop setup manual

Version 1.1 | Tested with CyberBrick Desktop 0.6.2 | Guide by Jonas_2603



Open the M1A2 Abrams SEP v3 RC Tank on MakerWorld

A beginner-friendly clean setup and recovery workflow using the CyberBrick Desktop application and the matching MakerWorld project. Screenshots reflect the desktop interface of the tested version; labels may change in later versions. Device names, numbers and project names in your own app will differ from the screenshots.

Using this guide with other CyberBrick models: this guide was written for the **M1A2 Abrams SEP v3** tank and its screenshots show that model. The workflow itself (registering the boards, updating firmware, assigning transmitter and receiver, sending the project and pairing) is the same for other CyberBrick RC models. Only the project file and its names change: always use the CyberBrick project provided with **your own model** (see Step 7).

Terms used in this guide: the **transmitter** is the board in the handheld remote control; the **receiver** is the board inside the tank. Every board is a CyberBrick controller core.

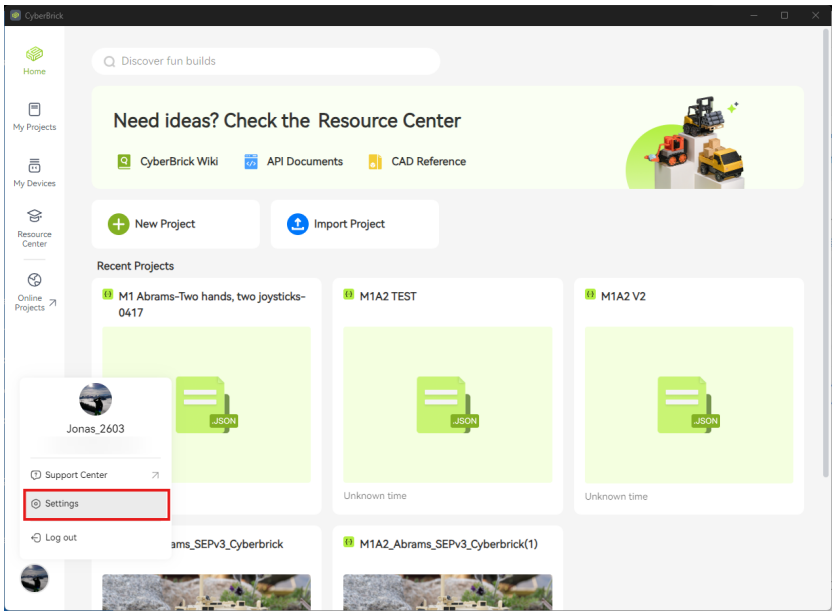
Before you begin: fully charge the correct batteries for the transmitter and the tank; use a known USB-C **data** cable (charge-only cables will not work); keep both boards accessible; keep the model secured for testing (for the tank: tracks off the ground). Never use an unsuitable battery or force a connector. Confirm model-specific wiring and power requirements in the assembly guide.

Do you need Steps 1–2? USB flashing is a clean-install/recovery path and is not necessarily required for every new board. If both boards already connect and accept the correct RC project, you may begin with Step 3. For a fresh clean setup or suspected incorrect firmware, use Steps 1–2.

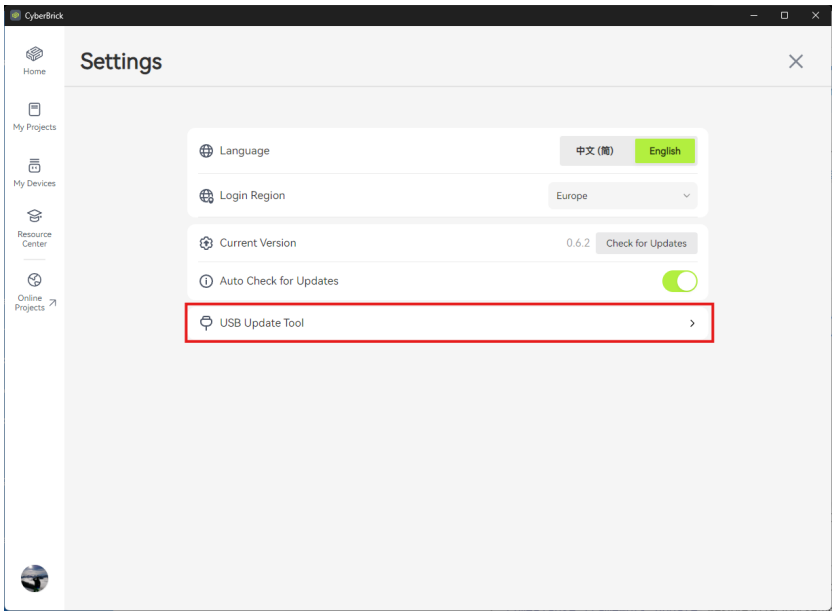
Connections at a glance: USB-C is used to register a board (Step 3), to update firmware (Step 4) and, if needed, for the clean-install flash (Step 2). Bluetooth is used from Step 5 onwards so CyberBrick Desktop can talk to the powered boards for configuration. Neither is needed while driving.

Step 1: Open the USB Update Tool

Launch CyberBrick Desktop. Open the profile menu on the Home screen, select **Settings**, then **USB Update Tool**.



Screenshot 1 — Open the profile menu and select Settings.



Screenshot 2 — Open USB Update Tool.

Step 2: Flash the RC framework (clean-install route)

Connect **only one controller core at a time** to the computer with a USB-C data cable. Follow the official wired firmware procedure exactly and, when the tool offers a choice, select the **RC framework** for CyberBrick. Wait for a clear success message before unplugging. Repeat for the second board.

Official instructions: Bambu Lab wired firmware guide. This guide is authoritative for the flashing buttons and sequence; do not guess recovery key combinations or use unrelated ESP32 firmware.

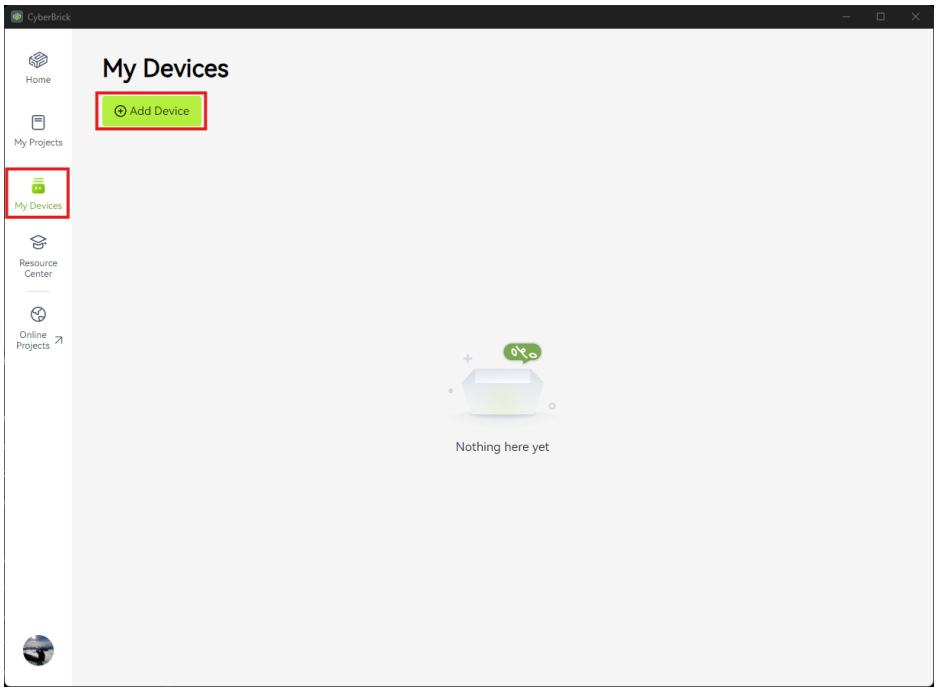
> **Safety:** never unplug the USB-C cable or interrupt power while firmware is being written. If flashing fails, retry only according to the official recovery instructions.

Step 3: Add the transmitter and receiver to My Devices

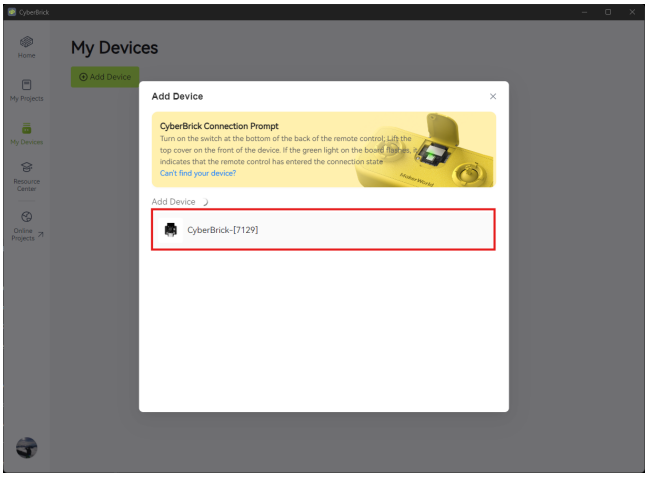
Connect the first board individually via USB-C. In CyberBrick Desktop, open **My Devices** → **Add Device**. Select the detected board and complete any prompts. Repeat for the other board, again connecting it individually.

If a board does not show up, follow the on-screen *CyberBrick Connection Prompt* (visible in Screenshot 4): for the remote, turn on the switch at the bottom of the back, lift the top cover on the front, and check that the green light on the board flashes.

Make a note of which physical board is the **transmitter (remote)** and which is the **receiver (tank)**.



Screenshot 3 — Open My Devices and choose Add Device.



Screenshot 4 — Choose the board detected in the Add Device dialog.

Step 4: Check for firmware updates

For each registered board, allow the application to finish connecting and run any offered **official** firmware updates. Wait until the board reports that it is up to date, then disconnect USB-C. Do not assume both boards have the same firmware status; check each one separately.

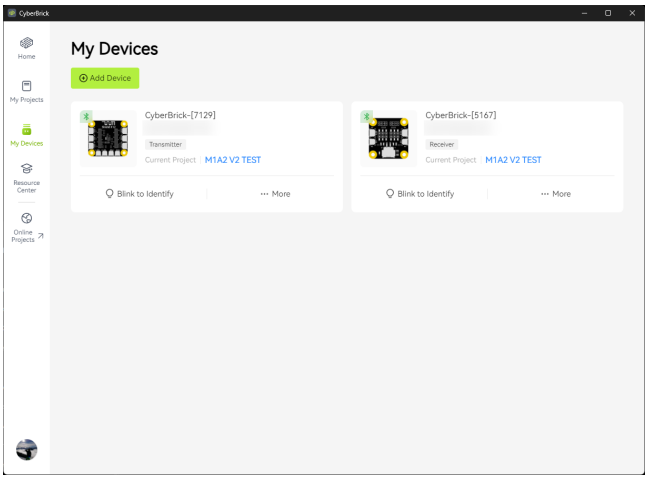
Step 5: Power both boards and connect by Bluetooth

Power the transmitter and receiver using their **correct, charged** batteries. Enable Bluetooth on your computer and keep both boards nearby and accessible.

USB-C was used to register and update the boards; from here on, Bluetooth connects each board to CyberBrick Desktop for configuration. Bluetooth to the computer is **not** the connection that must stay open while driving.

Step 6: Verify both boards in My Devices

Open **My Devices** again and connect both boards. Verify that both appear and are available before opening the project. If a board is missing, check power, Bluetooth permissions, and whether another device is already connected to it.



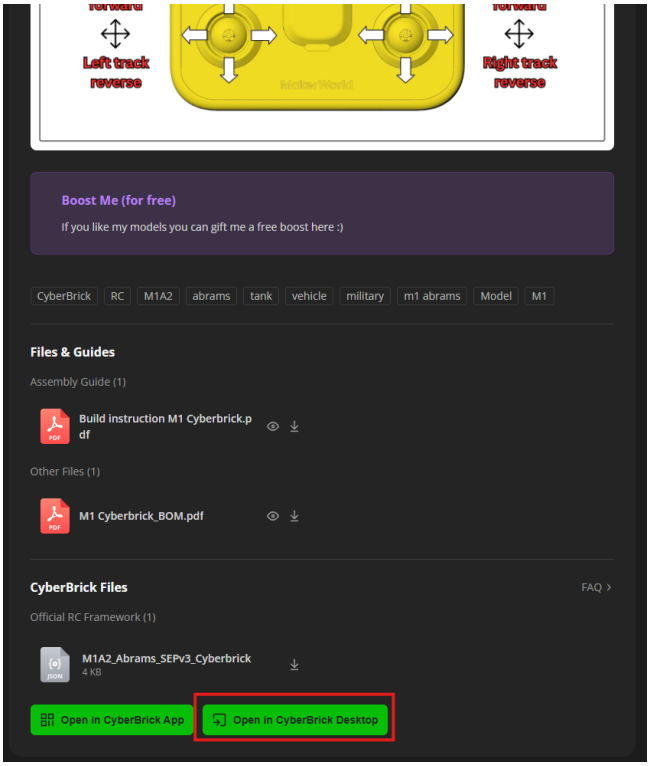
Screenshot 5 — Both registered boards shown under My Devices. (Your project names will differ.)

Step 7: Open the exact project for your model from MakerWorld

Go to the MakerWorld page of **your model**. For the tank this guide was written for, that is the **M1A2 Abrams SEP v3 CyberBrick** page (link). Under **CyberBrick Files**, click **Open in CyberBrick Desktop**.

Using a different CyberBrick model? Open that model's own MakerWorld page and use the CyberBrick file listed there. The remaining steps work the same way; only the project and receiver names will differ from the screenshots.

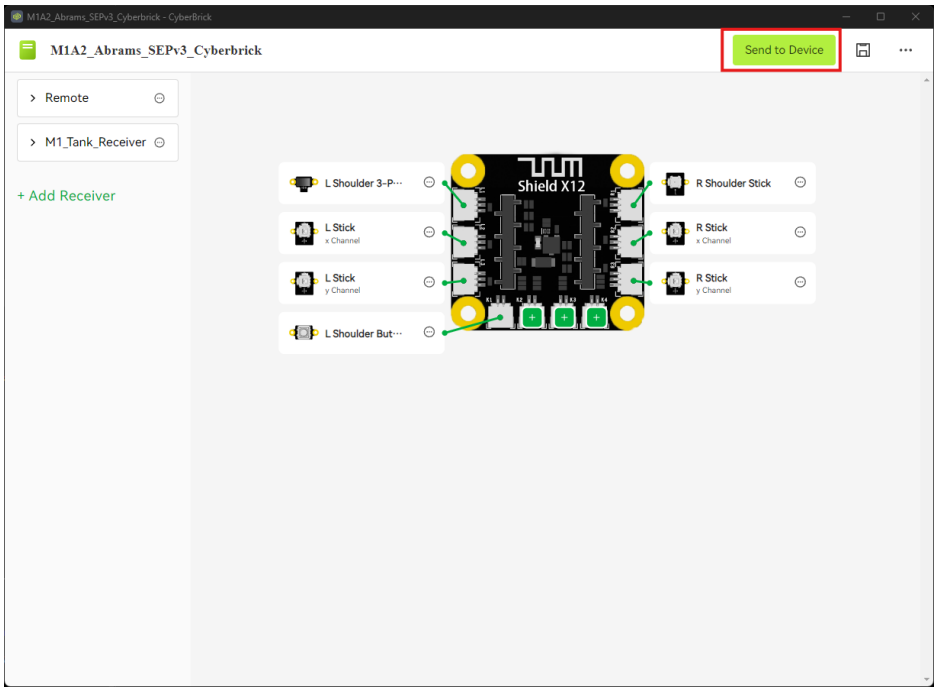
Always use the configuration provided with **your exact model**, not one from another model or another version of the same model. A wrong configuration can make motors or servos behave incorrectly.



Screenshot 6 — On MakerWorld, choose Open in CyberBrick Desktop.

Step 8: Open the project

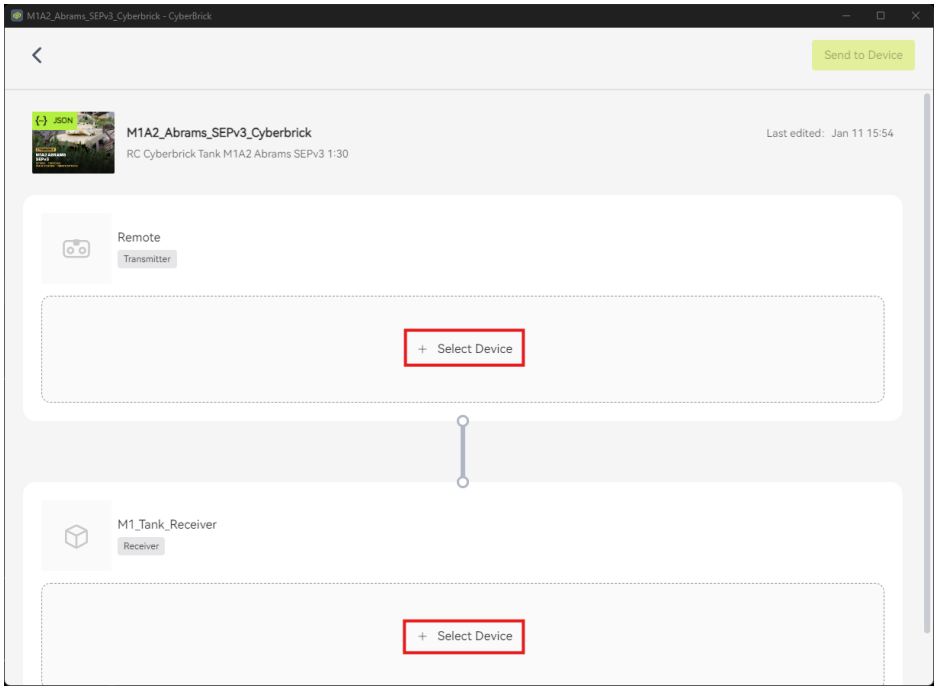
CyberBrick Desktop opens the project configuration. Check that it contains a **Remote** and a receiver (for the Abrams tank: **M1_Tank_Receiver**; other models show their own receiver name). Click **Send to Device** – this opens the device assignment page (the upload itself starts in Step 11).



Screenshot 7 — Open the project and click Send to Device.

Step 9: Select the remote (transmitter)

On the assignment page, click **+ Select Device** under **Remote / Transmitter**, then continue with Step 10 to choose your board.

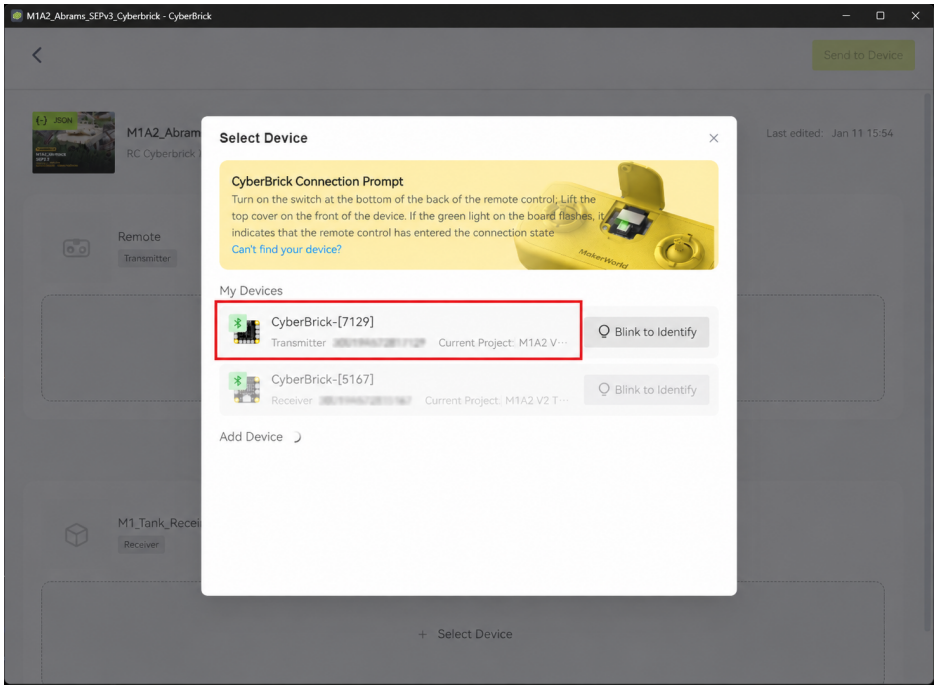


Screenshot 8 — Choose Select Device for transmitter and receiver.

Step 10: Assign the correct boards

Choose your transmitter board as **Transmitter**. The receiver may populate automatically; otherwise click **+ Select Device** under the receiver entry (**M1_Tank_Receiver** for the Abrams) and choose the receiver board.

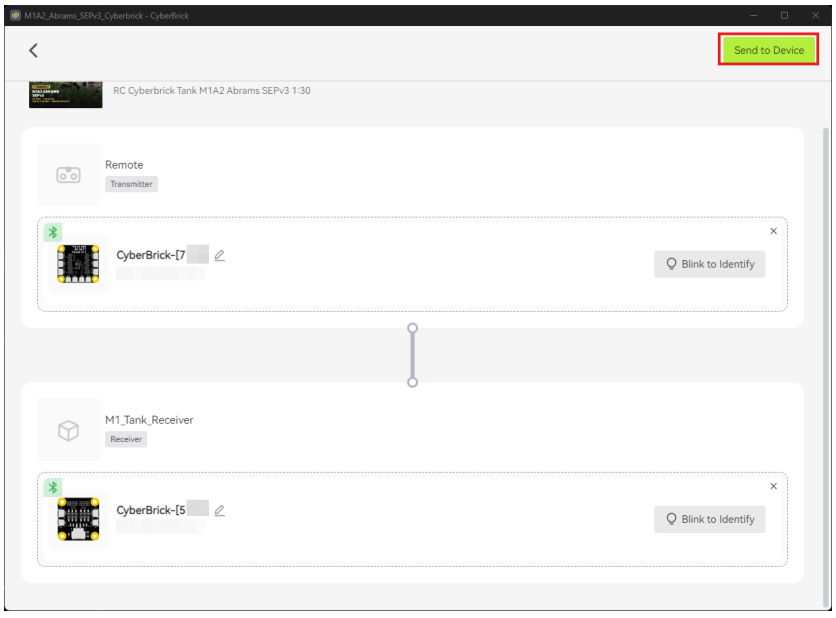
Double-check both assignments. Use **Blink to Identify** if you are uncertain which physical board is which. Do not assign the same board to both roles. If the transmitter does not appear in the list, return to Steps 5–6 and check power and Bluetooth.



Screenshot 9 — Select the correct transmitter board from My Devices.

Step 11: Send the configuration and pair

Once the correct transmitter and receiver appear in their slots, click **Send to Device**. This uploads the project and includes the pairing operation.

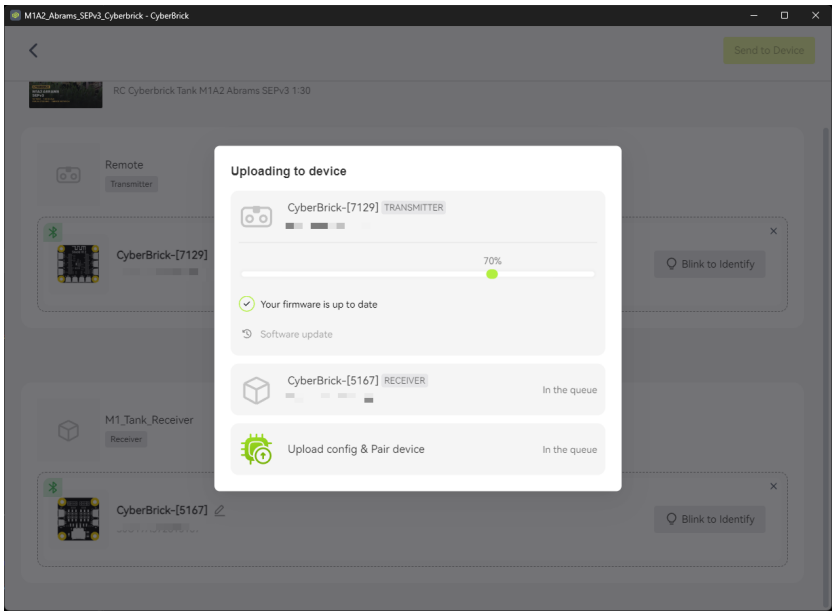


Screenshot 10 — Check both boards are assigned, then click Send to Device.

Step 12: Wait for every operation to finish

Leave both devices powered and connected. The software may process transmitter firmware/software, receiver firmware/software, and **Upload config & Pair device** in sequence. This can take several minutes.

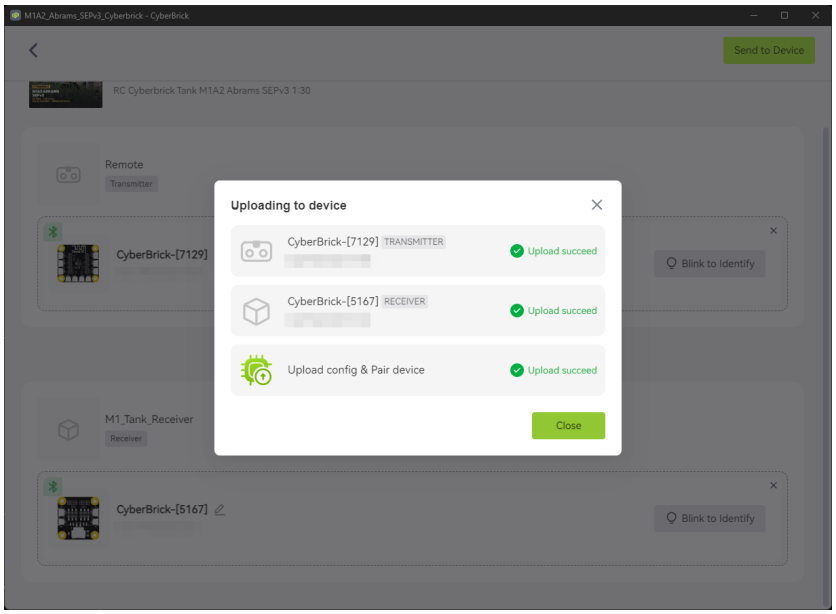
Do not disconnect Bluetooth or battery power while the process is running and do not switch off either board. A progress percentage on one board is not confirmation that the entire job is finished.



Screenshot 11 — Wait while firmware, software and pairing/configuration upload complete.

Step 13: Confirm success, then perform a safe test

Check for green success indicators for **Transmitter**, **Receiver**, and **Upload config & Pair device**. Only then click **Close**.



Screenshot 12 — All three operations report Upload succeed.

First test

Place the model so that nothing can drive away (for the tank: on a stand with both tracks clear of the surface). Switch on the transmitter and the tank, wait until both have started, and gently test each function (for the tank, both tracks and the turret). Release the sticks to confirm the motors stop. If controls behave unexpectedly, power down immediately and check the configuration and wiring. For the meaning of the status LEDs, see the official CyberBrick User's Manual (linked below).

Once the transmitter and receiver are paired and running their uploaded project, the computer and CyberBrick Desktop do not need to remain connected for normal RC operation.

Quick troubleshooting

- **Board not found (USB):** use a data-capable USB-C cable (not charge-only), try another port, and connect only one board at a time.
- **Board not found (Bluetooth):** make sure the board is powered by a charged battery, Bluetooth is enabled with permission granted to the app, and the board is not already connected to another device. For the remote, check the green light is flashing (see the Connection Prompt in Screenshot 4).
- **Wrong device assignment:** use Blink to Identify, then reselect transmitter and receiver.
- **Upload interrupted:** recharge batteries, reconnect both boards, and retry from the application; do not improvise low-level flashing steps.
- **Both uploads succeeded but the tank does not move:** check power supply, motor wiring, model-specific receiver ports, and RC pairing. Test with the drive wheels or tracks raised.
- **Receiver USB-C becomes inaccessible after assembly:** complete the setup and first functional test before permanently enclosing it.

References and verification notes

Official wired flashing instructions:
https://wiki.bambulab.com/en/cyberbrick/troubleshooting/wired-firmware-guide

Official Maker's Supply CyberBrick User's Manual:
https://3d.nice-cdn.com/upload/file/Maker_s_Supply_CyberBrick_User_s_Manual.pdf

Bambu Lab community desktop-app discussion:
https://forum.bambulab.com/t/cyberbrick-desktop-app-manual-instructions-anything/220577